Intrinsic Evaluation of Word Embeddings

Team Members:

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Part 1 — Intrinsic Evaluation of Word Embeddings: Glove vs word2vec(skip-gram) of size 300

syntactic Analogy Tests

Test #	# Test	Expected Answer	GloVe 300 Result	Word2Vec 300 Result	Fasttext300 Results
1	run:running::swim:?	swimming	swimming	swimming	swimming
2	govern:government::achieve:?	achievement	achieved	achieving	achieves
3	mouse:mice::goose:?	geese	geese	geese	geese
4	happy:happier::heavy:?	heavier	heavier	heavier	heavier
5	quick:quickly::slow:?	slowly	slowly	slowly	slowly
6	fast:fastest::slow:?	slowest	slowest	slowest	slowest
7	write:writer::teach:?	teacher	teaches	taught	teacher
8	create:creation::relate:?	relation	relates	relates	relates
9	nation:national::culture:?	cultural	cultural	cultural	cultural
10	music:musical::history:?	historical	historical	Broadway_musical	historical

Semantic Analogy Tests

Test #	# Test	Expected Answer	GloVe 300 Result	Word2Vec 300 Result	Fasttext300 Results
1	paris:france::rome:?	italy	italy	italy	germany
2	apple:fruit::spinach:?	vegetables	vegetables	fresh_spinach	vegetables
3	tokyo:japan::ottawa:?	canada	canada	canadian	canada
4	japan:yen::india:?	rupee	rupees	rupee	rupee
5	camera:photo::mic:?	sound	someșul	mike	mike
6	doctor:hospital::teacher:?	school	school	elementary	school
7	light:dark::day:?	night	days	night	night
8	car:driver::airplane:?	pilots	plane	plane	pilots
9	seed:plant::caterpillar:?	butterfly	factory	plants	Caterpillar
10	dawn:sunrise::dusk:?	sunset	sunset	sunset	sunset

Analogy Accuracy

Analogy Type	GloVe 300 Accuracy	Word2Vec 300 Accuracy	Fasttext300 Accuracy
Semantic	50.00%	40.00%	70.00%
Syntactic	60.00%	50.00%	70.00%

Part 2 — Intrinsic Evaluation of Word Embeddings: Glove vs fastText of embedding size 100.

Semantic Analogy Tests

Test #	# Test	Expected Answer	GloVe 300 Result	GloVe 100 Result	GloVe 50 Result
1	paris:france::rome:?	italy	italy	italy	italy
2	apple:fruit::spinach:?	vegetables	vegetables	lettuce	lettuce
3	car:driver::airplane:?	pilots	plane	pilot	pilot
4	japan:yen::india:?	rupee	rupees	rupees	rupees
5	camera:photo::mic:?	Sound	Someșul	Gazette	ajc
6	doctor:hospital::teacher:?	School	school	School	school
7	light:dark::day:?	Night	Days	Days	couple
8	seed:plant::caterpillar:?	butterfly	Factory	Motors	nummi
9	tokyo:japan::ottawa:?	canada	canada	Canada	finland
10	dawn:sunrise::dusk:?	Sunset	Sunset	Sunset	sunset

syntactic Analogy Tests

Test #	# Test	Expected Answer	GloVe 300 Result	GloVe 100 Result	GloVe 50 Result
1	run:running::swim:?	Swimming	Swimming	Swimming	swimming
2	govern:government::achieve:?	Achievement	Achieved	Progress	support
3	mouse:mice::goose:?	Geese	Geese	cows	calves
4	happy:happier::heavy:?	Heavier	Heavier	Heavier	heavier
5	quick:quickly::slow:?	Slowly	Slowly	Rapidly	rapidly
6	write:writer::teach:?	Teacher	teaches	Teacher	sociologist
7	develop:development::argue:?	Argument	Argued	Argued	argued
8	create:creation::relate:?	Relation	Relates	Relates	relates
9	nation:national::culture:?	Cultural	Cultural	Cultural	cultural
10	music:musical::history:?	Historical	Historical	Historical	historical

Analogy Accuracy

Analogy Type	GloVe 300 Accuracy	GloVe 100 Accuracy	GloVe 50 Accuracy
Semantic	50.00%	40.00%	30.00%
Syntactic	60.00%	50.00%	40.00%

Analysis

Part 1:

Performance Overview:

Semantic Analogy

1. **fastText's Superior Performance (70%)**

- Successfully solved 7 out of 10 semantic analogies
- Correctly handled: paris→italy, apple→vegetables, car→pilots, japan→rupee, doctor→school, light→night, dawn→sunset
- Failed on: camera→mic (predicted "mike" instead of "sound"), seed→caterpillar (predicted "Caterpillar" the brand), tokyo→ottawa (predicted "canada" instead of maintaining the pattern)

2. **GloVe's Moderate Performance (50%)**

- Successfully solved 5 out of 10 semantic analogies
- Strong on geographic and conceptual relationships: paris→italy, apple→vegetables, japan→rupees, tokyo→canada, dawn→sunset
- Failed on domain-specific analogies: camera→mic, seed→butterfly, light→day

3. **Word2Vec's Lower Performance (40%)**

- Successfully solved 4 out of 10 semantic analogies
- Correct on: paris→italy, japan→rupee, light→night, dawn→sunset
- Struggled with compound concepts: apple→"fresh_spinach", car→"plane", doctor→"elementary", seed→"plants"

Syntactic Analogy

1. **fastText's Consistent Excellence (70%)**

- Successfully solved 7 out of 10 syntactic analogies
- Perfect on: run→swimming, mouse→geese, happy→heavier, quick→slowly, write→teacher, nation→cultural, music→historical

- Failed on morphological transformations: govern→achievement (got "achieves"), develop→argument (got "arguing"), create→relation (got "relates")

2. **GloVe's Strong Syntactic Performance (60%)**

- Successfully solved 6 out of 10 syntactic analogies
- Excellent on simple transformations: run→swimming, mouse→geese, happy→heavier, quick→slowly, nation→cultural, music→historical
- Struggled with derivational morphology: govern→achievement, write→teacher, develop→argument, create→relation

4. **Word2Vec's Moderate Performance (50%)**

- Successfully solved 5 out of 10 syntactic analogies
- Strong on basic transformations: run→swimming, mouse→geese, happy→heavier, quick→slowly, nation→cultural
- Failed on complex derivations and word formations

fastText

- Uses character n-grams, enabling understanding of word parts (prefixes, suffixes).
- Handles morphology well (e.g., *run → running*, *rupee → rupees*).
- Can represent out-of-vocabulary words by combining subword vectors.
- Produces more robust embeddings for syntactic and rare words.

GloVe

- Learns from global co-occurrence statistics, not just local windows.
- Stable and strong for common and frequent words.
- Captures semantic and syntactic patterns from corpus-wide statistics.
- Balanced performance across tasks, but less morphological awareness.

Word2Vec

- Uses local context windows only
- Lacks subword modeling → struggles with morphology and rare words.
- Performance depends heavily on training corpus coverage.
- Weaker at capturing complex morphological relationships.

Similarities and Differences

Similarities:

1. Basic Analogies: All three models successfully handled simple, well-established analogies:
   - Geographic: paris→rome, tokyo→ottawa
   - Temporal: dawn→dusk
   - Morphological basics: run→swim, happy→heavy
2. Consistent Ranking: fastText > GloVe > Word2Vec held true for both semantic and syntactic tasks
3. Difficulty Patterns: All models struggled with:
   - Abstract or unusual semantic relationships (camera→mic)
   - Complex derivational morphology (govern→achievement)
   - Less common word pairs

Differences:

1. **Performance Gap Magnitude**:
   - **Semantic Task**: Large gap between models (30% spread: 40% to 70%)
   - **Syntactic Task**: Smaller gap (20% spread: 50% to 70%)

- **Interpretation**: Syntactic regularities are more systematic and easier to capture; semantic relationships are more varied and context-dependent

2. **Model-Specific Strengths**:
   - **fastText**: Consistently high on both tasks due to subword modeling
   - **GloVe**: Better on syntactic (60%) than semantic (50%), suggesting global co-occurrence captures syntactic patterns well
   - **Word2Vec**: Similar performance gap but lower absolute performance

3. **Error Patterns**:
   - **Word2Vec errors**: Often retrieves related but incorrect words (e.g., "fresh_spinach" instead of "vegetables")
   - **GloVe errors**: Sometimes gets different forms (e.g., "rupees" instead of "rupee") or unrelated words
   - **fastText errors**: Occasionally finds morphologically similar words that don't fit semantically (e.g., "Caterpillar" the brand)

Training Mechanisms:

GloVe — Global Co-occurrence Matrix Factorization

- Builds a word-word co-occurrence matrix from the entire corpus
- Factorizes the matrix to produce word embeddings
- Preserves global co-occurrence statistics
- Combines global statistical information + local context
- Optimizes embeddings to reconstruct co-occurrence probabilities

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fastText — Subword Modeling Using Character N-grams

- Extends Skip-gram by breaking each word into character n-grams
- Represents each word as the sum of its subword (n-gram) vectors
- Learns embeddings for character sequences
- Generates embeddings even for out-of-vocabulary words

- Embedding construction uses subword composition instead of full-word only

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Word2Vec (Skip-gram) — Local Context Prediction

- Learns embeddings by predicting surrounding words from the target word
- Uses only the local context window
- Trains through predictive learning, not matrix factorization
- Embeddings depend solely on contextual co-occurrences within window
- No use of subword information or global statistics

strengths and weaknesses

Model	**Best Use**	**Limitation**
fastText	**Morphology, rare words**	**Bigger & heavier**
GloVe	**Global semantic structure, efficiency**	**No rare words or morphology**
Word2Vec	**Speed & scalability**	**local-only context**

Part 2:

Quality:

- Higher dimensional embeddings (e.g., GloVe-300) provide richer and more detailed word representations, allowing the model to capture subtle semantic, syntactic, and contextual differences.
- Medium dimensions (GloVe-100) retain core meanings but compress information, losing fine-grained distinctions.
- Low dimensions (GloVe-50) suffer from severe information loss, leading to confusion between unrelated words and weak relational understanding.

Analogy Accuracy:

Syntactic relationships consistently perform 10% better across all dimensions, suggesting syntactic structure is more compressible and systematic than semantic meaning

Dimension	Semantic Accuracy	Syntactic Accuracy
300d	50%	60%
100d	40%	50%
50d	30%	40%

Trade-off Between Dimensionality, Computational Cost, and Performance

Higher Dimensions (e.g., 300d)	Lower Dimensions (e.g., 50d)
Higher accuracy and richer representations	Lower accuracy and weaker understanding
Better analogy reasoning	More errors and nonsensical outputs
Higher computational cost—more memory & slower training	Lower computational cost—faster & lighter
Suitable for complex NLP tasks	Suitable only when speed/storage matters